MILITARY SPECIFICATION

ANTENNA ASSEMBLY AS-2049/VPS-2: 10548388

1. SCOPE

1.1 This specification covers an Antenna Assembly AS-2049/VPS-2 which is used as the antenna for the Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacturing and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of
MIL-S-50666	Servo Drive, Traverse: 10548052
MIL-S-50667	Servo Drive, Elevation: 10548062

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10548388	Antenna Assembly, AS-2049/VPS-2
D10548193	Servo Electronics
C10549134	Connector, Plug, Miniature
F10549103	Base Support Assembly

PACKAGING DATA SHEET

10548388	Packaging of Antenna Assembly

(Copies of specifications, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. - The antenna assembly shall be manufactured in accordance with Drawing F10548388 and drawings pertaining thereto. The subassemblies of the antenna shall have been inspected and accepted in accordance with their respective drawings or specifications. The following subassemblies are covered by the indicated specifications:

UNIT NO.	NAME	SPECIFICATION
A2	Servo Drive, Elevation: 10548062	MIL-S-50667
A3	Servo Drive, Traverse: 10548052	MIL-S-50666

3.2 General specifications. - Where applicable, the antenna assembly shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerance
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. - The antenna assembly shall be mounted on the vibration table in normal service position. The antenna assembly shall meet the requirements of 3.1 and 3.5 after being subject to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 40 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

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3.3.2 Shock. - The antenna assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 ±1 milliseconds (ms). The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 ms.

3.3.3 Altitude.

3.3.3.1 Operating. - The antenna assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. - The antenna assembly shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. - The antenna assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40° F and +155° F.

3.3.4.2 Nonoperating (storage). - The antenna assembly shall not suffer permanent damage when subjected to not less than 4 hours at each temperature extreme of -65° F and +160° F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). - The antenna assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 95 ±5 percent for a period of not less than 10 days.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the antenna assembly.

3.5 Performance. - The antenna assembly shall meet the radio frequency characteristics when tested on an antenna test range which shall provide far field vision $\left(\frac{2D^2}{\lambda}\right)$ This requires a minimum of 75 feet between transmitting and receiving antennas for the measurements.

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3.5.1 Antenna pattern. - The side lobe energy of any side lobe shall be at least 22 decibies (dB) below the main lobe energy.

3.5.2 Beamwidth. - The beam width of the main lobe shall be 3.4° to 4.0° for energy received.

3.5.3 Gain. - The antenna assembly shall have a gain of at least 31 dB over an isotropic radiator.

3.5.4 Boresight. - The optical axis of the boresight clamp and the electrical center of the antenna beam shall be within ±1/8°.

3.5.5 VSWR. - The VSWR of the antenna assembly, when pointed in a nonreflecting direction, shall not be greater than 1.4:1.

3.5.6 Servo drive dynamic test. - The elevation and traverse servo drives shall be driven by servo electronics, 10548193, with synchro input signal which causes the antenna to oscillate sinusoidally ±25°.

3.5.6.1 When the elevation and traverse servos are driven at 0.1 Hz, the antenna shall track to less than 1.5 milliradians (mrad). The antenna performance shall be such that the "DEMOD" output signal (TP3) of the appropriate servo electronics, 10548193, when filtered to the 10 to 30 Hz passband and converted to a true root mean square (rms) signal and recorded as shown in Figure 1, shall not exceed 30 millivolts (mV). For the recorded waveforms shown in Figure 1, Channel 1 displays the filtered "DEMOD" output signal while channel 2 displays the filtered and averaged signal of Channel 1 so as to provide a less complex waveform for quantitive analysis. Figure 1 shows (a) an acceptable antenna, (b) a nonacceptable antenna having a peak dc output exceeding 30 mV, and (c) a nonacceptable antenna having, in addition to greater than 30-mV level, visible vibrations present in the reflector and feed assembly.

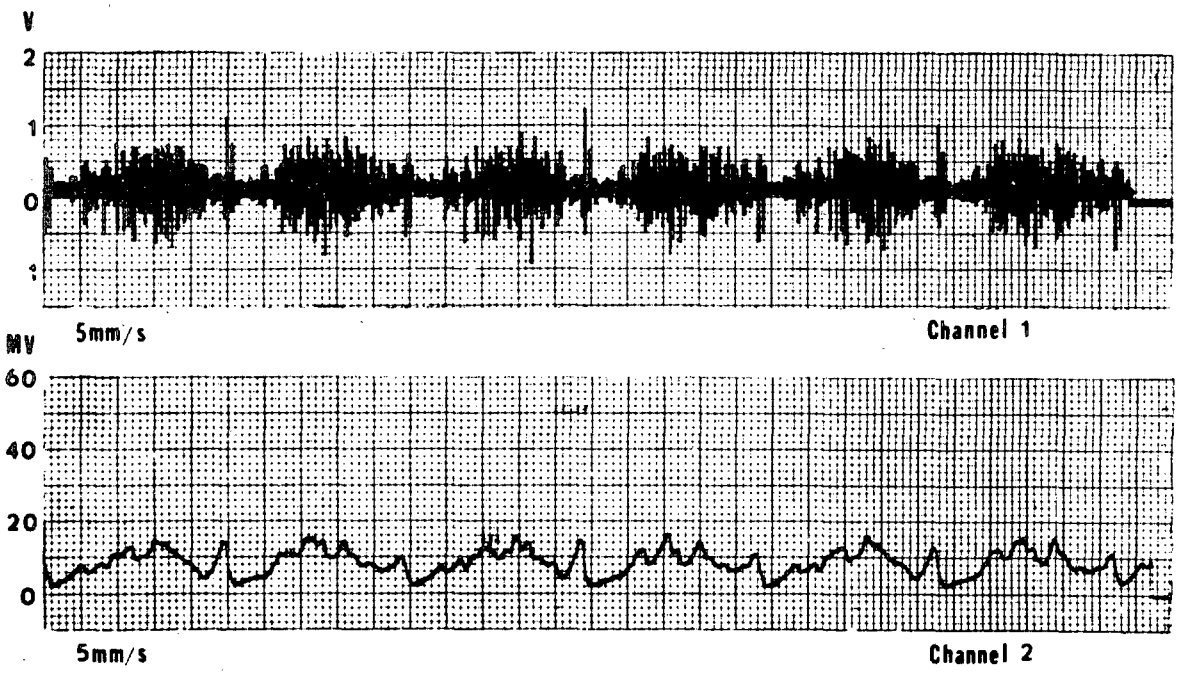
3.5.7 Function potentiometers.

3.5.7.1 Elevation. - With 15 volts direct current (Vdc) applied to J1-g and J1-f of the elevation servo drive, the λ^2E potentiometer output at J1-S shall provide the following voltages using J1-h as dc return:

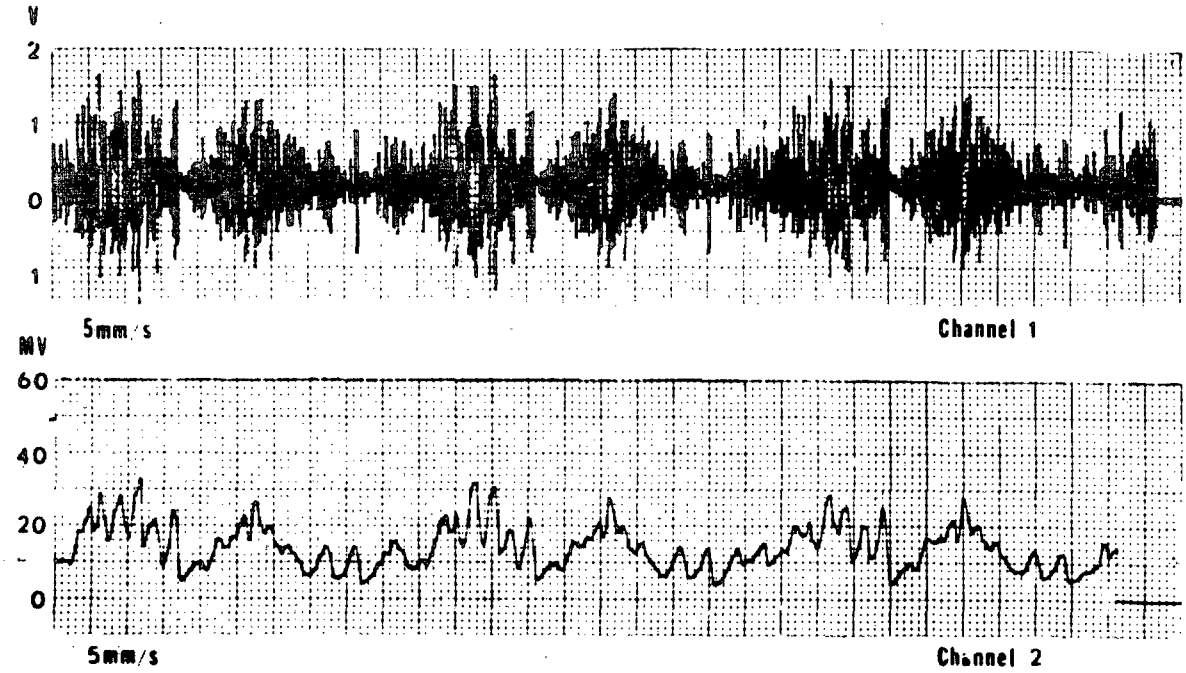
a. A null of not more than 120 mV at 0° elevation lead angle (trunnion output).

b. A 2.4 ±0.4 V at ±10° elevation lead angle (±10° lead angle corresponds to ±12.25° of synchro input information).

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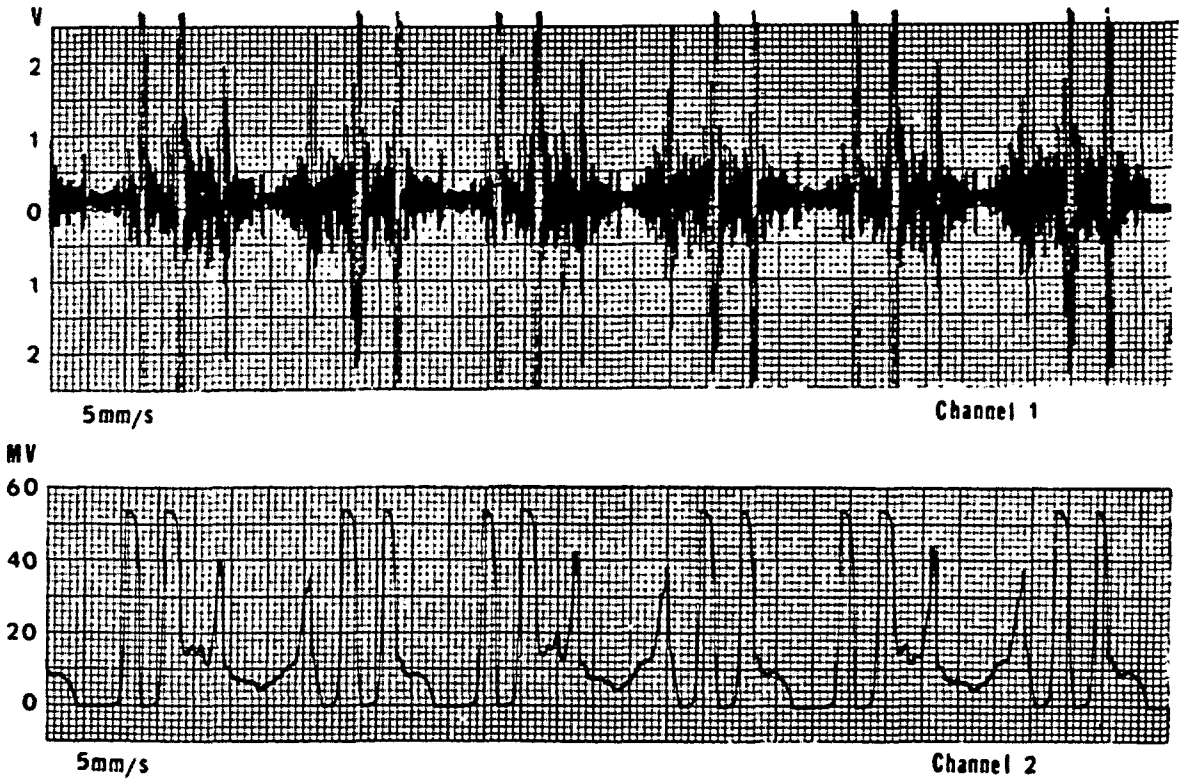
(a) Acceptable



(b) Non – acceptable

FIGURE 1. CHART RECORDING DISPLAYS. (sheet 1 of 2)

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(c) Non-acceptable, visible vibration

FIGURE 1. CHART RECORDING DISPLAYS (sheet 2 of 2)

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c. The difference between the +10° and -10° lead angle voltages shall not exceed 0.2 V.

3.5.7.2 Traverse. - With 15 Vdc applied to J1-g and J1-f of the traverse servo drive, the λ^2_T potentiometer output at J1-s shall provide the following voltages using J1-h as dc return:

a. A null of not more than 120 mV at 0° traverse lead angle (trunnion output).

b. 2.4 ±0.4 volts at ±10° traverse lead angle (±10° lead angle corresponds to ±12.7° of synchro input information.)

c. The difference between the +10° and -10° lead angle voltages shall not exceed 0.2 V.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) submission. - The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three antenna assembly manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contractor. The sample and a record of the inspection, including certification for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.

4.3.2.1 Component and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. - Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. - The requirements and tests in Table I shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (60° F and 90° F).

TABLE I. - CLASSIFICATION OF DEFECTS
(100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE		
MAJOR:		
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3
MINOR: NONE		

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Special sampling. - One antenna assembly shall be selected at random by a Government representative as a control sample from each 100 produced . The control sample shall meet the requirements and tests in Table II. Three antenna assemblies shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table III.

TABLE II. - SPECIAL SAMPLING

	CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings - Visual
302.	General specifications	3.2	MIL-F-13926 & MIL-C-46357 - Visual

TABLE III. - SPECIAL SAMPLING, ENVIRONMENTAL

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude	3.3.3	4.6.1.3
306.	Temperature	3.3.4	4.6.1.4
307.	Humidity (nonoperating)	3.3.5	4.6.1.5

4.4.1 Failure of sample - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment. - Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

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4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in TDP.

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment. - In conjunction with 4.5.2.2, the standard test equipment listed in Table IV shall be utilized in performing the applicable tests of 4.6.

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TABLE IV. - TEST EQUIPMENT LIST

ITEM NUMBER		DESCRIPTION
1.	Signal generator	Frequency: 9.15 to 9.25 GHz. Rf power output: 1 and 100 mW. Modulator: AM modulation to 50% at 1 kHz, tunable ±10%.
2.	SWR meter and detector	Sensitivity: 0.15 μV, tuned to 1 kHz; calibrated to read SWR with a square-law detector.
3.	Slotted line and detector	Frequency: 9.15 to 9.25 GHz. Residual SWR: Less than 1.05:1. Sensitivity: sufficient to provide greater than 1 μV at 1 kHz. Square law over greater than 20 dB.
4.	Attenuator	Attenuation: 10 dB SWR: 1 to 1:1 Frequency: 9.2 GHz
5.	Variable attenuator	Attenuation: 0 to 22 dB ±5%. Frequency: 9.2 GHz.
6.	Directional coupler	Coupling factor: 20 dB ±1 dB. SWR: 1 to 1:1.
7.	Frequency meter	Frequency: 9.15 to 9.25 GHz. Accuracy: ±1%.
8.	Standard gain horn	Gain between 15 and 25 dB at 9.2 GHz.
9.	Antenna test range	Distance between antennas: 75 ft Std. antenna gain: sufficient to record VPS-2 antenna pattern, with minimum 30 dB dynamic range. Pedestal repeatability: ±1 min Pedestal azimuth travel: 360° Pedestal elevation travel: ±20° from horizon Frequency: 9.15, 9.2 & 9.25 GHz. Recorder: Signal vs. direction ±1 min.

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TABLE IV. - TEST EQUIPMENT LIST (Cont'd).

ITEM NUMBER		DESCRIPTION
		Visual boresight: Target displaced in same direction from center of transmitting antenna by equal amount that boresight clamp is displaced from center of antenna assembly. (11.25 ±.01 in.)
10.	Test fixture (optional)	Fixture to mount antenna and return antenna in both planes, to within ±1 min of center between two electrical measurements.
11.	Servo control unit	See Figure 2. See Table V for parts list.
12.	Dc power supply	24 Vdc, 10 A. Load regulation: ±1%. Ripple: less than 100 mV.
13.	Dc power supply	15 Vdc, 100 mA. Load regulation: ±0.1% Ripple: less than 10 mV.
14.	Ac-dc voltmeter	Range: 500 mV to 150 V. Input impedance: 10 megohms.
15.	Shaft clamp	See Figure 3.
16.	Servo director	Synchro transmitter compatible with 11CT4e. Simple harmonic motion of one cycle per 5 and 10 seconds with ±25° adjustable sector scan.
17.	Rms voltmeter with dc output	Response: Responds to rms value (heating value) of input signal. Frequency range: 10 Hz to 10 mHz. Fullscale accuracy: ±5% from 10 Hz to 50 Hz. Voltage range: 0.3 V to 3 V fullscale.

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TABLE IV. - TEST EQUIPMENT LIST (Cont'd)

		Dc output: Ac to dc converter. Accuracy: ±5% for 10 Hz to 50 Hz. Output: 1 Vdc into open circuit for full scale deflection, proportional to meter deflection; 1 ma maximum: Nominal source impedance 1000Ω. Input impedance: 10 MΩ shunted by 50 PF. Output noise: 1 mV rms.
18.	Band pass filter	Frequency response: 0.02 Hz to 20 kHz. Pass-band gain: 0 dB ±1 dB. Attenuation slope: nominal 24 dB per octave. Maximum attenuation: greater than 80 dB. Input impedance: 22 megohms or greater. Parallel capacitance: 200 ±50 pF.
19.	Servo amplifier	In accordance with Frankford Arsenal drawing D10548193.
20.	Recorder	Two channel. Sensitivity: 5mV/cm to 20 V/cm. Speed: 1.0 to 100 mm/s Frequency response: 80 Hz.
21.	Power source, 400 Hz	115 V, 1 A with distortion less than 1%.
22.	Base support assembly	In accordance with Frankford Arsenal drawing F10549103.

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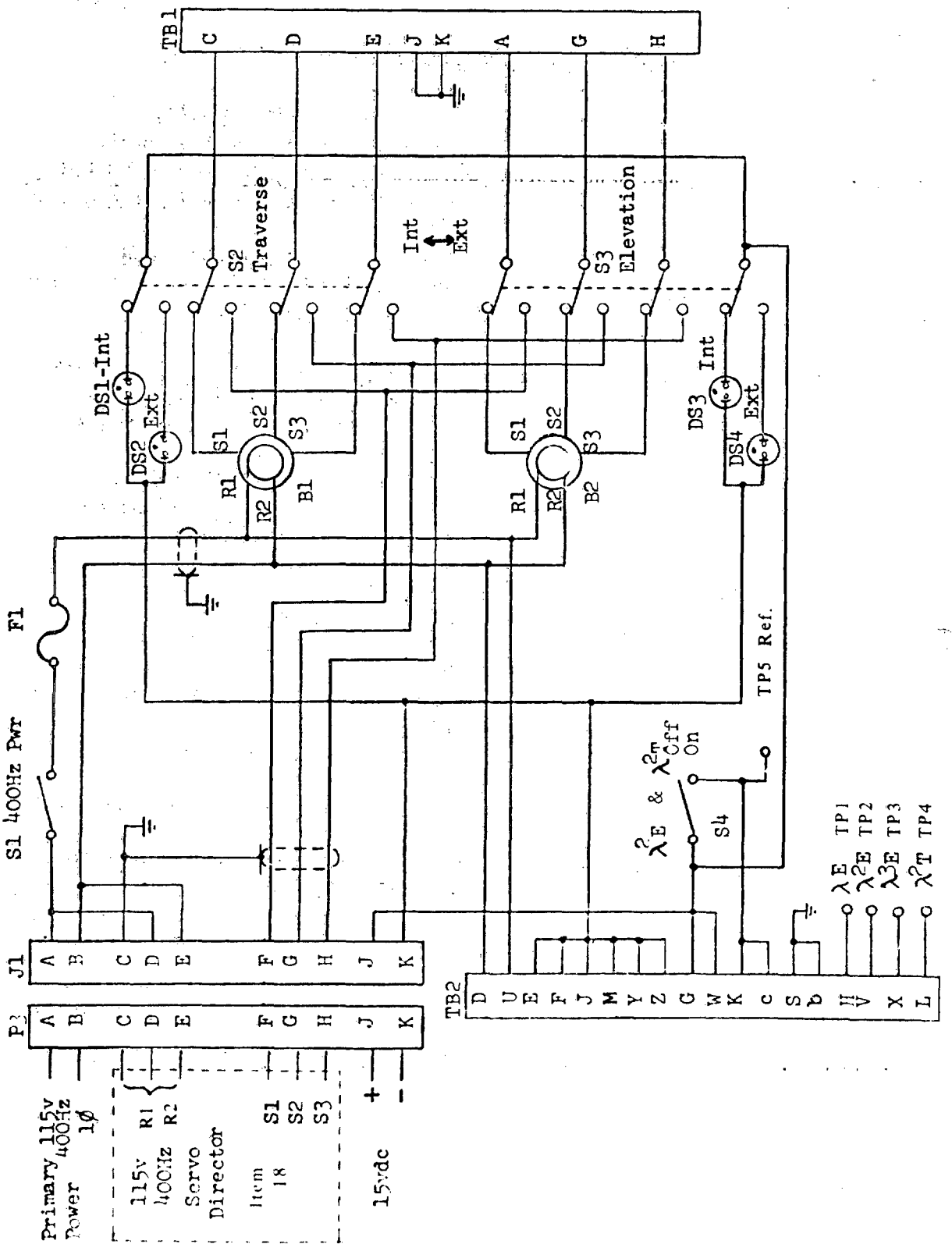


FIGURE 2. SERVO CONTROL UNIT

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TABLE V. - SERVO CONTROL UNIT PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
J1	KPT02A-20-16P, chassis mounting
P1	KPT06B-16-26s, Cannon
TB1	Terminal board
TB2	Terminal board
B1	Synchro control transmitter, with dial
B2	Synchro control transmitter, with dial
S1, S4	Switch, SPST, 2A
S2, S3	Switch, 4PDT, 15A
F1, XF1	Fuse holder, and fuse, 1A
DS1 thru DS4	Neon glow lamp, NE-51
XDS1 thru XDS4	Lamp holder, Dialco 502-8136-102
TP1 thru TP5	Tip jack, H.H. Smith - 1501 or equivalent.

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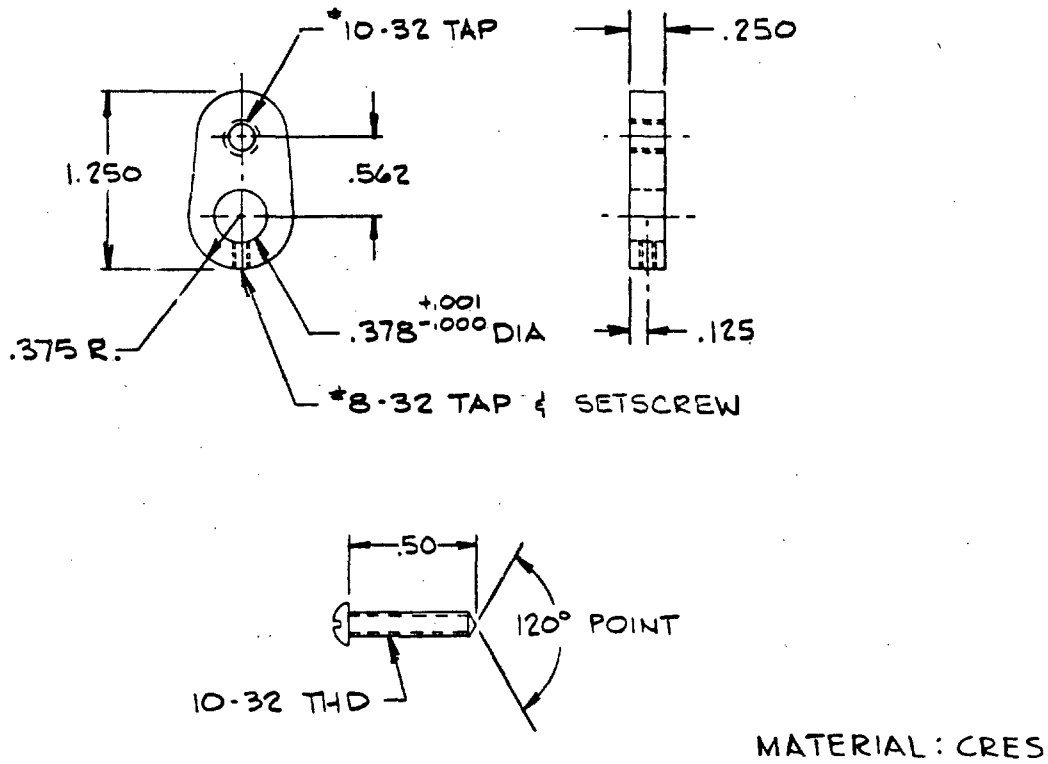


FIGURE 3. SHAFT CLAMP

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4.6 Test methods and procedures.

4.6.1 Environmental. - Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration. - The test shall be performed as specified in Method 514, Procedure IX, Part I, of MIL-STD-810 except that failure criteria do not apply. After vibration, the antenna assembly shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.1.

4.6.1.2 Shock. - This test shall be performed as specified in Procedure 1 of Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms and 15 G. After the shock impulses, the antenna assembly shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.2.

4.6.1.3 Altitude (nonoperating and operating). - This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 feet (5.5 inches Hg). Operation of Step 3 shall be the performance tests specified in 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-810. The temperature of Step 2 shall be +160° F. The temperature of Step 3 shall be maintained for not less than 4 hours. The temperature of Step 4 shall be +155° F. In Step 5 the temperature shall be maintained for 4 hours and the antenna assembly shall be tested in accordance with 4.6.2 and 4.6.3 during this time.

4.6.1.4.2 Low (operating and nonoperating). This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -65° F maintained for not less than 4 hours. The temperature of Step 4 shall be -40° F. In Steps 5 and 7, the antenna assembly shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity (nonoperating). - This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810. The antenna assembly shall be tested in accordance with 4.6.3.

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4.6.2 Electrical. - All components and subassemblies shall have been electrically checked for continuity at a lower level of assembly.

4.6.3 Performance. - The selected antenna assemblies shall be tested as follows: The tests for gain, boresight, VSWR, servo drive, dynamic test, and function potentiometers (4.6.3.3, 4.6.3.4, 4.6.3.5, 4.6.3.6, and 4.6.3.7) shall be performed on all antenna assemblies produced. The tests for antenna pattern (4.6.3.1) and beamwidth (4.6.3.2) need only be performed on the first article and at least 10 percent of production equipment. The tests for gain, boresight, antenna pattern and beamwidth (4.6.3.3, 4.6.3.4, 4.6.3.1, and 4.6.3.2) may be performed on the reflector and feed subassembly (drawing 10548028) at the discretion of the contractor.

4.6.3.1 Antenna pattern. - The antenna pattern is to be obtained as follows:

a. Connect signal generator, item 1, to antenna test range transmitting antenna (part of item 8) through variable attenuator, item 9, and adjust frequency for 9.2 gigahertz (GHz).

b. Use test fixture, item 9, to mount antenna assembly to pedestal and receiver of antenna test range (part of item 8) and point antenna assembly toward test transmitting antenna. Adjust antenna test range equipment controls for a peak on test range recorder while turning, in all directions, antenna assembly under test for maximum recorder deflection.

c. Adjust test range equipment for full recorder deflection.

d. Rotate antenna under test to 180° (in azimuth) from transmitting antenna and make a 360° azimuth recording of antenna assembly under test. Return antenna to peak deflection position of antenna test range and increase variable attenuator setting by 22 dB.

e. Note the recorder deflection. Decrease variable attenuator setting by 22 dB. Rotate the antenna to all side lobes. Note maximum recorder deflection for each side lobe. The side lobes of the antenna assembly as recorded above shall not be greater than the 22 dB down reference level.

f. Repeat a. to obtain full deflection. Rotate antenna under test in a vertical direction to 20° from horizontal. Record pattern in elevation ±20° from horizon. Return antenna to peak deflection of recorder and increase variable attenuator attenuation by 22 dB.

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g. Note the recorder deflection. Decrease variable attenuator setting by 22 dB. Rotate the antenna vertically to all side lobes. Note maximum recorder deflection for each side lobe. The side lobes of the antenna assembly as recorded above shall not be greater than the 22 dB down reference level.

h. Repeat a. through g. for frequencies 9.15 and 9.25 GHz.

4.6.3.2 Beamwidth. - The beamwidths will be obtained by repeating 4.6.3.1 a. for 9.2 GHz.

a. Operate recorder to obtain expanded directional pattern (i.e.,6° to 60°).

b. Start azimuth pattern at least 3° deflection from peak obtained above and record at least 3° past peak.

c. Return antenna assembly to peak and adjust variable attenuator to obtain 3 dB attenuation.

d. Use this level to mark antenna pattern with 3 dB reference.

e. The 3 dB beamwidth as obtained shall be between 3.4° and 4.0°.

f. Repeat a. through e. except rotation will be in elevation.

g. The 3 dB beamwidth as obtained shall be between 3.4° and 4.0°.

h. Repeat a. through h. for frequencies of 9.15 and 9.25 GHz.

i. The 3 dB beamwidth as obtained shall be between 3.4° and 4.0°.

4.6.3.3 Gain. - The gain shall be obtained by performing 4.6.3.1 a. for 9.2 GHz using antenna test range or optional test fixture (no recording required).

a. Adjust the antenna test range equipment to obtain a variable attenuation of a least 3 dB greater than 31 dB, minus the "standard gain horn" gain when full recorder deflection or similar indication is obtained.

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b. The antenna assembly under test is replaced with standard gain antenna and variable attenuator adjusted to obtain full recorder deflection.

c. The change in variable attenuator setting plus the gain of the standard gain antenna shall be no less than 31 dB.

4.6.3.4 Boresight. - The boresight of the antenna shall be accomplished using the antenna test range (item 9) or holding fixture which allows movement of the antenna, ±4° from center and return to center within ±1 minute from electrical measurement.

a. The antenna shall be adjusted for electrical center (peak) by repeating 4.6.3.1 a. for 9.2 GHz using antenna test range or test fixture.

b. Adjust variable attenuator to increase attenuation by 3 dB and record level of recorder deflection.

c. Return variable attenuator to value before 3 dB increase.

d. Rotate antenna assembly until recorded level of 3 dB above is passed in azimuth.

e. Return antenna assembly back to obtain 3 dB reference above (above operations to overcome backlash).

f. Note reading or indication and continue rotation through peak and to next 3 dB reference. Note reading.

g. Move antenna assembly to halfway between the two 3 dB readings taking care to go past desired reading and return so approach is from same direction as before.

h. Repeat d. for elevation. Place telescope in boresight telescope holder and observe boresight target in cross hairs.

4.6.3.5 VSWR. - Standing wave measurements are obtained by connecting signal generator, item 1, output to slotted line and detector, item 3, with attenuator, item 4, between item 1 and 3.

a. Connect SWR meter, item 2, to detector output.

b. Connect output of slotted line to antenna assembly under test and aim antenna in a nonreflecting direction.

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c. Turn equipment on and allow to stabilize.

d. Adjust frequency of signal generator to 9.2 GHz.

e. Adjust the modulation frequency of signal generator to obtain a peak reading of the SWR meter.

f. Adjust the signal generator attenuator level and slotted line probe for a full scale reading on the SWR meter and so that all measurements are made with in the square law region of the slotted line detector.

g. Obtain the SWR as indicated on the SWR meter by movement of the slotted line probe for maximum to minimum voltage levels.

h. Repeat e. for 9.15 and 9.25 GHz. SWR shall be less than 1.4 to 1.

4.6.3.6 Servo drive dynamic test. - The servo drives shall be tested as follows:

a. Connect the antenna assembly to the test equipment as shown in Figure 4. Connect the servo electronics, item 19, to W1 and filter as shown by the solid outline in Figure 4, for the performance test on the elevation servo drive of the antenna.

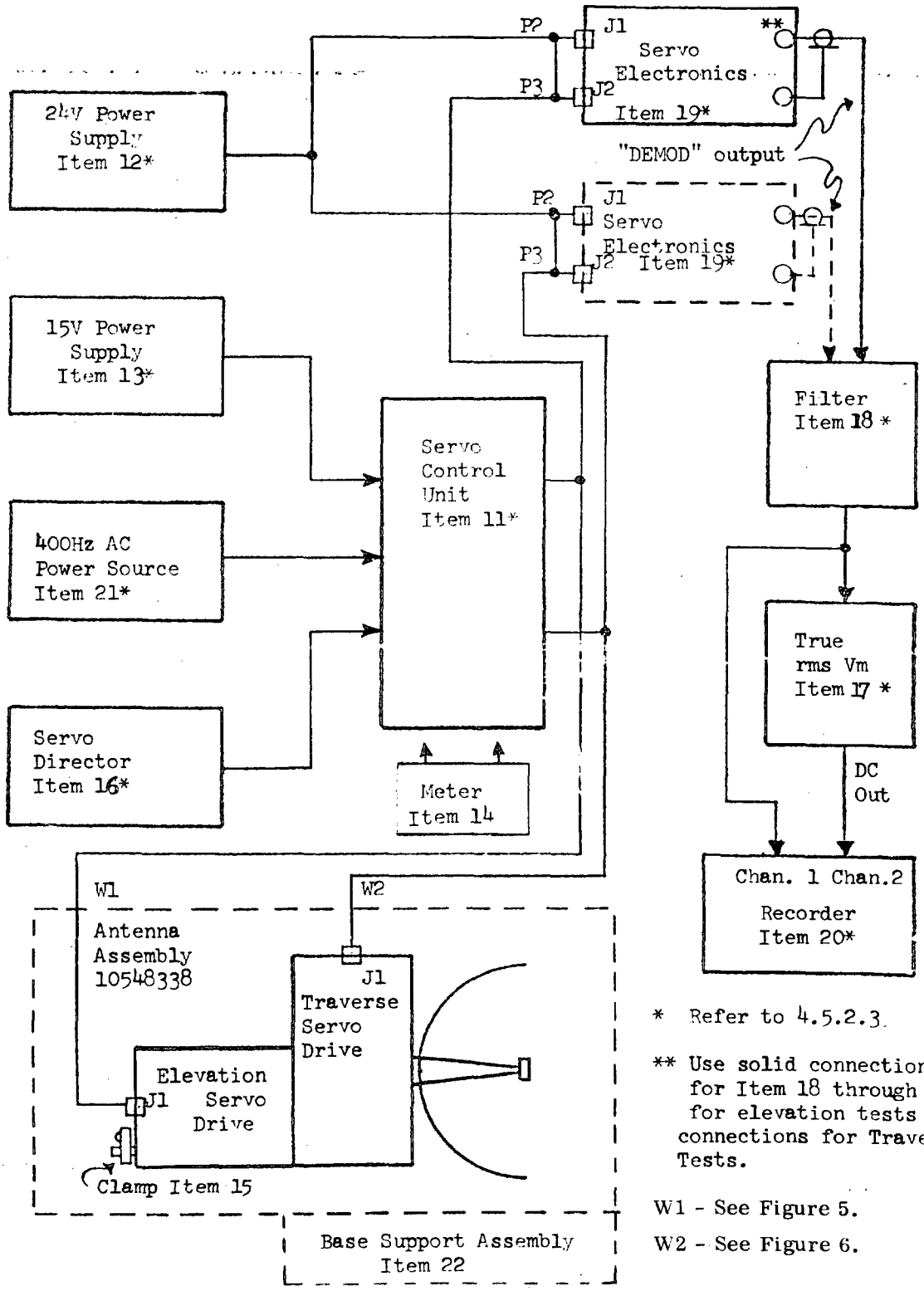
b. The antenna assembly shall be mounted to a rigid platform during the following tests. Install the shaft clamp, item 15, to the elevation drive shaft and with the servo control unit, item 11, in the internal mode of operation set the elevation dial to 0°. Apply power to the antenna and adjust the shaft so the antenna is approximately 25° above the horizontal, then lock in place with the shaft clamp.

c. Set the servo control unit to the external mode and adjust the servo director so that it is oscillation 25° either side of 0° at a 0.1 Hz rate.

d. Set the recorder, item 20, to operate at a chart speed of 5 millimeters per second (5 mm/s); Channel 1 gain at 1 volt per centimeter (1 V/cm) and Channel 2 gain at 20 millivolts per centimeter (20 mV/cm). Establish a zero reference for Channel 1 at the center of the chart and a zero reference for Channel 2 near the edge of the chart (See Figure 1).

e. Set the filter, Item 18, for a band pass of 10 Hz to 30 Hz.

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* Refer to 4.5.2.3.

** Use solid connections for Item 18 through 20 for elevation tests and-- connections for Traverse Tests.

W1 - See Figure 5.

W2 - See Figure 6.

FIGURE 4. TEST SETUP.

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f. Set the true rms voltmeter, item 17, to the 1-volt scale.

g. With the antenna scanning ±25° of 0° reference record four cycles of antenna operation. The maximum value of the signal as recorded on Channel 2 (rms voltmeter dc output) shall be 30 mV.

NOTE: In addition to the 30-mV limit being exceeded, the basic criteria for rejection of an antenna servo drive is chatter or vibration of the antenna dish or feed horn readily discernable by observation. Figure 1(c) displays a chart recording of such a condition.

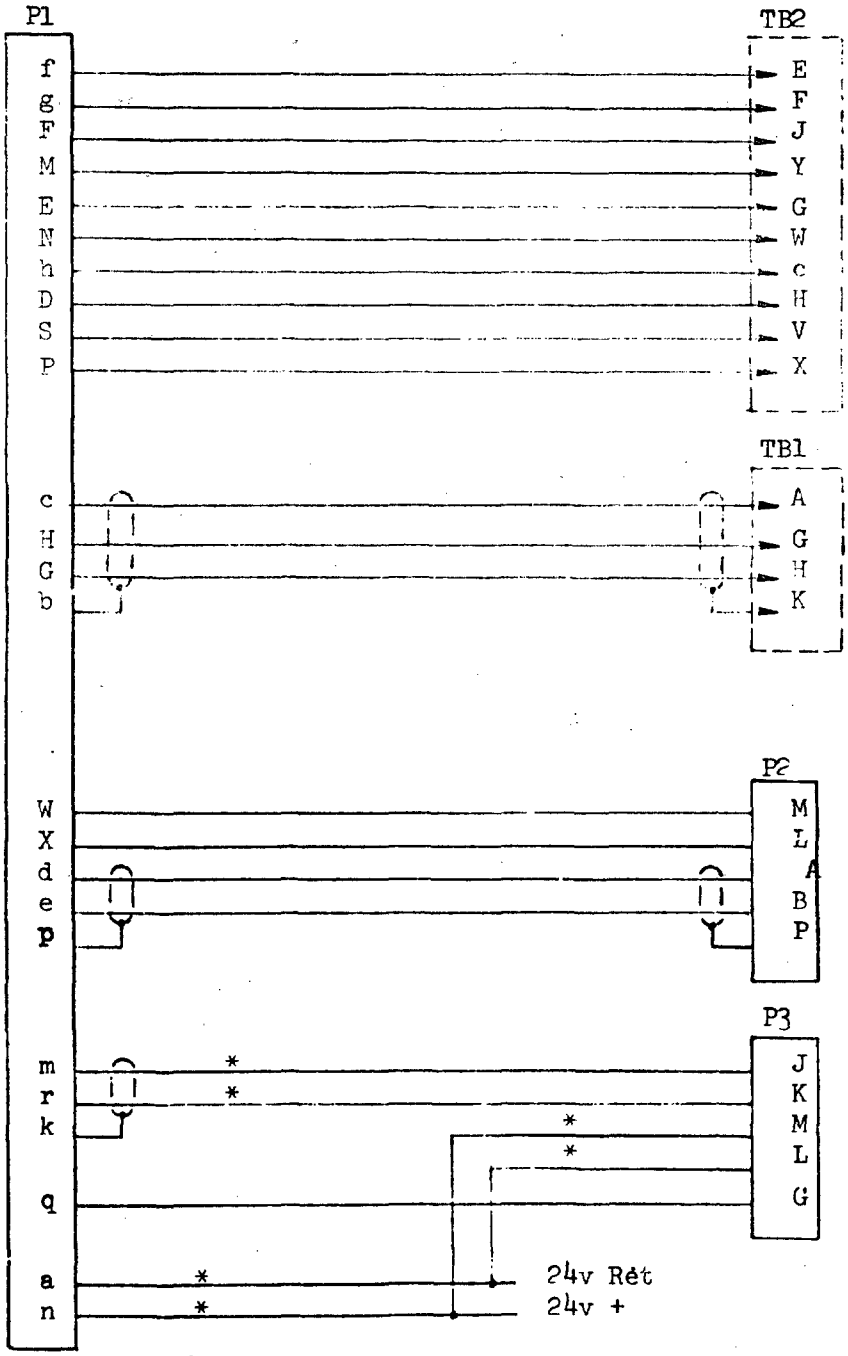
h. For the performance test on the traverse servo drive, connect the servo electronics, item 19, to W2 and perform steps c through g.

4.6.3.7 Function potentiometers. - For test of the function potentiometers set up antenna assembly as shown in Figure 4. Synchros B1 of each servo drive shall have been adjusted for electrical 0°. Operate the servo control unit, item 11, to the internal mode of operation, and turn S4 to the on position.

4.6.3.7.1 Elevation. - Using the voltmeter, item 14, at TP2 on the servo control unit measure the λ^2_E potentiometer output to be as specified in 3.5.7.1. Use TP5 on the servo control unit as reference for voltage measurements. When the lead angle is specified as ±10°, position the dial of the servo control unit to ±12.25°.

4.6.3.7.2 Traverse. - Using the voltmeter at TP4 on the servo control unit measure λ^2_T potentiometer output to be as specified in 3.5.7.1. Use TP5 on the servo control unit as reference for voltage measurements. When the lead angle is specified as ±10° position the dial of the servo control unit to ±12.7°.

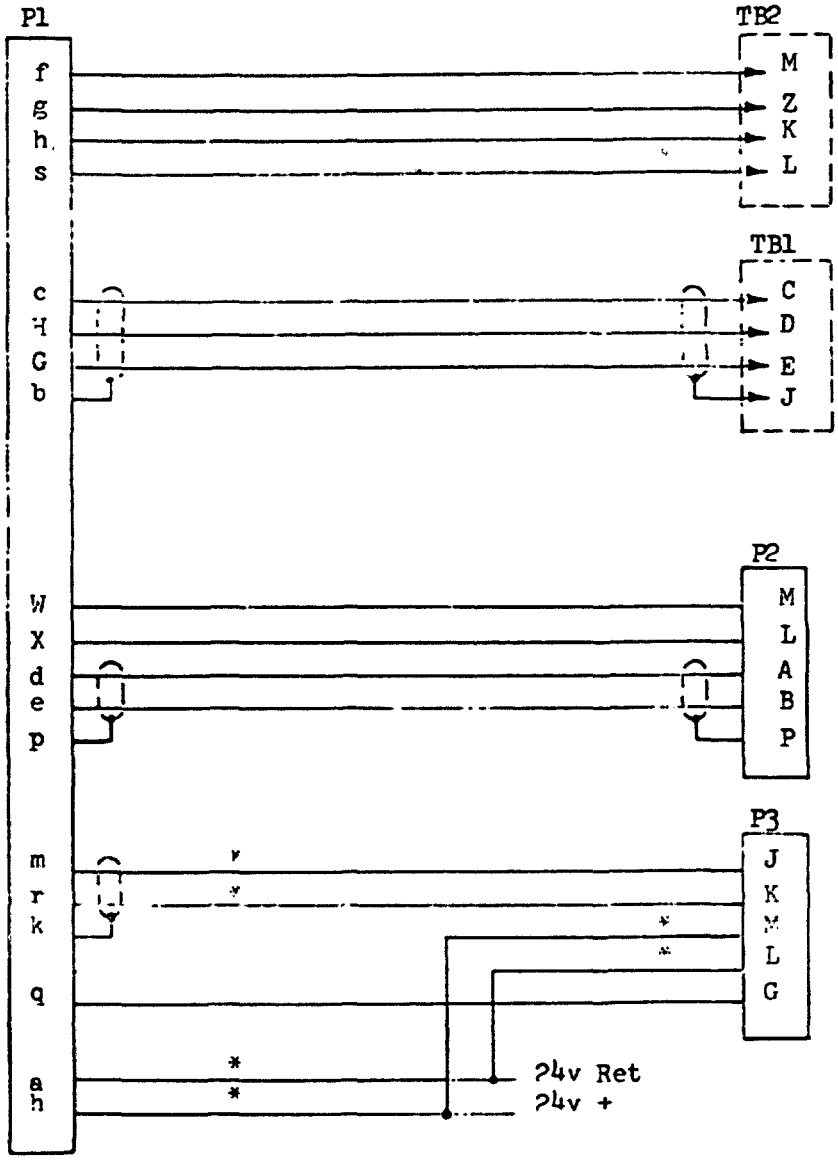
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* Conductor size 16 AWG, all others 22 AWG.
P1 - Connector, plug Frankford Arsenal part no. 10549134-8
P2 - Connector, plug - MS3116E-14-19-S
P3 - Connector, plug - MS3116C-14-12-S
TB1 & TB2 - part of Servo Control Unit Item 11.

FIGURE 5. TEST CABLE W1.

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* Conductor size 16AWG, all others 22 AWG

P1 - Connector, plug Frankford Arsenal part no. 10549134-7

P2 - Connector, plug - MS3116E-14-19-S

P3 - Connector, plug - MS3116E-14-12-S

TB1 & TB2 part of Servo Control unit item 11.

FIGURE 6. TEST CABLE W2.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10548388. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The antenna assembly is mounted on a vehicle and is used as an antenna for Radar Set AN/VPS-2. In this function, it transmits pulsed rf power and receives radar echoes. Serco positioning of the antenna permits up to ±25° offset in traverse from the position of the associated weapon and elevation positioning from -50° to +95°.

6.2 Ordering data. - Procurement documents should specifiy the following:

a. Title, number, and date of this specification

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Block of serial numbers to identify each antenna assembly.

g. Requirement for first article submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285-A290

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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